

Bala Kumaran M

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PROFESSIONAL SUMMARY

Enthusiastic and results-driven Computer Science & Engineering student with hands-on experience in Java Full-Stack Development and Blockchain Technology. Proven track record of hackathon victories, research publications, and leadership as Google Gemini Student Lead. Proficient in building decentralized systems, scalable web applications, and automated testing.

TECHNICAL SKILLS

- **Languages:** Java, Python, Solidity, JavaScript, HTML, CSS
- **Frameworks & Libs:** Spring Boot, React.js, Node.js, Express.js, Three.js, AR.js
- **Tools & Databases:** Truffle, Ganache, Git, GitHub, Docker, Google Cloud (GCP), MySQL, MongoDB, Firebase, Power BI

PROFESSIONAL EXPERIENCE

Blockchain Developer Intern — Technohacks (1 Month (2024))

- Researched decentralized protocols and wrote custom smart contracts using Solidity.
- Utilized Truffle and Ganache to configure local blockchain environments and deploy contracts.
- Wrote Mocha/Chai unit tests to validate state transitions and variables of smart contracts.

Java Full Stack Intern — i5 Technology (1 Month)

- Developed responsive web application features using Java, Spring Boot, and Hibernate/JPA mapping.
- Created REST APIs and mapped structured data to MySQL schemas with input check rules.

Data Analytics Intern — Cognifyz Tech (Remote) (1 Month)

- Built data cleaning and preparation scripts using Python (Pandas and NumPy).
- Implemented K-Means clustering and keyword sentiment visualization with Seaborn and Matplotlib.

KEY PROJECTS

SecureLandX (Decentralized Land Registry) | React.js, Solidity, Ethereum, Flask, MetaMask, Tesseract OCR

- Developed Solidity contracts to guarantee tamper-proof and immutable land registry ownership records.
- Integrated MetaMask validation with a Flask backend and automated Patta document OCR verification.

Blockchain-Based Voting System | React.js, Solidity, Ethereum, Truffle, MetaMask, Ganache

- Created a decentralized voting dApp with secure, role-based flows (Admin and Voter validation).
- Integrated MetaMask for voter verification and Ganache for executing transparent local voting logs.

Vehictronix 360 (AR/VR & IoT Troubleshooting) | AR.js, Three.js, Unity, Vuforia, ESP32, C#

- Designed an AR system overlaying 3D troubleshooting guides onto large vehicle telemetry dashboards.
- Configured ESP32 microcontrollers to stream real-time diagnostic telemetry using Three.js.

Movie Ticket Admin Portal | Spring Boot, Java, MySQL, REST API, MVC

- Built a Spring Boot administrative portal featuring session control and role-based access rules.
- Optimized MySQL database mappings for transaction logging and custom exception tracking.
- Other Projects: SychDesk (MERN Workspace), Hostel Management PWA, Student CRUD App (Spring Boot), Cognifyz Data Analysis, Blockchain Env Setup (Truffle)

EDUCATION

B.E. in Computer Science & Engineering — A.V.C. College of Engineering (2022 - 2026 | CGPA: 7.21/10)

HSC in Basic Mechanical Engineering — S.M.H. Higher Secondary School (2020 - 2022 | Grade: 73%)

HONORS & LEADERSHIP

- Google Gemini Student Ambassador: Google Student Lead representing AVC College of Engineering campus.
- Research Publications: Published on 'Blockchain Secure Land Registry' (ICICEA-2026) and 'Blockchain for AI' (ICRET-2025).
- Hackathon Awards: 1st Place (SDG Symposium 2025 Hackathon); 3rd Place (Horizon 8 TANSAM Hackathon with cash prize).